
J1A
nuMezzanine_4x34
TCLK
T00
T01
T02
T03
T04
T05
T06
T07
T10
T11
T12
T13
T14
T15
T16
T17
T20
T21
T22
T23
T24
T25
T26
T27
TX
TY
J1C
nuMezzanine_4x34
BCLK
B00
B01
B02
B03
B04
B05
B06
B07
B10
B11
B12
B13
B14
B15
B16
B17
B20
B21
B22
B23
B24
B25
B26
B27
BX
BY
J1E
nuMezzanine_4x34
VCC_TL
VCC_BL
VCC_BR
VCC_TR
3V3_SELECT
5V_SELECT
GND
VCC_TL
GND
J1B
nuMezzanine_4x34
LCLK
L00
L01
L02
L03
L04
L05
L06
L07
L10
L11
L12
L13
L14
L15
L16
L17
L20
L21
L22
L23
L24
L25
L26
L27
LX
LY
J1D
nuMezzanine_4x34
RCLK
R00
R01
R02
R03
R04
R05
R06
R07
R10
R11
R12
R13
R14
R15
R16
R17
R20
R21
R22
R23
R24
R25
R26
R27
RX
RY
DATA BUS
U1A
uPD70337L-8-xxx
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
MULTIPLEXED ADDRESS / BUS CONTROL
U1B
uPD70337L-8-xxx
A0
A9/A1
A10/A2
A11/A3
A12/A4
A13/A5
A14/A6
A15/A7
A16/A8
A17/A9
A19
A18/UBE
IOSTB
MREQ
MSTB
R/W
REFRQ
PORT 0
U1C
uPD70337L-8-xxx
P00
P01
P02
P03
P04
P05
P06
P07/CLKOUT
PORT 1
U1D
uPD70337L-8-xxx
P10/NMI
P11/INTP0
P12/INTP1
P13/INTP2/INTAK
P14/INT/POLL
P15/TOUT
P16/SCK0
P17/READY
PORT 2 / DMA / ARBITRATION
U1E
uPD70337L-8-xxx
P20/DMARQ0
P21/DMAAK0
P22/TCO
P23/DMARQ1
P24/DMAAK1
P25/TC1
P26/HLDAK
P27/HLDRQ
SERIAL / RESET / CLOCK / MODE
U1F
uPD70337L-8-xxx
RxD0
CTSO/RSCK
RxD1
CTS1
N/S
X1
X2
RESET
TxD0
TxD1
PORT T / COMPARATOR REFERENCE
U1G
uPD70337L-8-xxx
PT0
PT1
PT2
PT3
PT4
PT5
PT6
PT7
VTH
POWER / NEC RESERVED-PIN BIAS
U1H
uPD70337L-8-xxx
VDD
VDD
IC
IC
IC
RESERVED
GND
GND
PWR_FLAG
GND
C1
100nF
C2
100nF
C3
100nF
At U1 pin 80
At U1 pin 50
At U1 pin 76
R1
6.8k
R2
15k
NEC reserved pin 9: 5–10k to GND
NEC IC pins 3,49,75: shared 10–20k to VDD
HOST REQUIREMENTS
5V_SELECT is directly grounded: 3V3_SELECT is floating.
Drive valid inputs before DUT power-on: RESET is active LOW.
X1 and X2 require complementary phases of the same clock.
GPIO ports switch in compatible groups: see bank-modes.md.
Reserved-pin resistors follow NEC 4G-2: no bank-input pulls.
Suffix xxx is the unknown factory opcode translation-table code.
R3
47k
R4
47k
R5
47k
R6
47k
R7
47k
R8
47k
B0: B00, B01
B2: B23, B24, B25, B26, B27
R0: R00, R01, R02
R2: R24, R25, R26, R27
T0: T00, T01, T02, T03, T04, T05
T1: T10, T11, T17
UNUSED PINS IN USED BANKS: ONE SHARED 47k PULL-DOWN PER BANK
Drive all tied unused bits LOW together whenever host -> DUT, including startup.
See unused-bank-pins.csv for connector pad numbers.
uPD70337L-8-xxx / PLCC84 socket / 5 V / DMA and bus arbitration
Sheet: /
File: v35s.kicad_sch
Title: NEC V35S DUT daughter board
Size: A2
Date: 2026-09-13
Rev: 1.0
KiCad E.D.A. 10.0.6-rc2
Id: 1/1